

COVER PAGE

What to Include

- University name
- College name
- Department name
- Project title
- Authors
- Advisor
- Co-advisor (if any)
- Submission date
- Place

Writing Rule

Use the exact approved title:

Real-Time Facial Emotion Detection Desktop Application Using Deep Learning

Do not make the title longer.

DECLARATION PAGE

Purpose

Confirms originality.

Content

- Student declaration
- Signature lines
- Date

Length

1 page

APPROVAL PAGE

Include

- Advisor approval
- Examiner approval
- Department approval
- Signatures

Length

1 page

ACKNOWLEDGMENT

Length

1 page

Structure

- Paragraph 1
 - Thank God.
- Paragraph 2
 - Thank advisor.
- Paragraph 3
 - Thank department.
- Paragraph 4
 - Thank university.
- Paragraph 5
 - Thank friends and family.

Writing Style

Formal.

Avoid emotional language.

ABSTRACT

Length

250–350 words

Structure

Paragraph 1 – Problem

Why emotion recognition matters.

Paragraph 2 – Objective

What the study aims to achieve.

Paragraph 3 – Methodology

Mention:

- Dataset aggregation
- Five model generations
- Three application generations
- CNN
- Transfer learning
- Knowledge distillation
- ONNX deployment

Paragraph 4 – Results

Mention:

- Final model accuracy (72.13%)
- Real-time desktop application
- ONNX Runtime deployment
- YuNet face detection

Paragraph 5 – Contribution

Mention:

- Iterative FER optimization framework
- Practical deployment framework
- Ethiopian implementation context

Keywords

5–7 keywords

Example:

- Facial Emotion Recognition
- Deep Learning
- Knowledge Distillation
- ConvNeXt
- EfficientNetV2
- ONNX Runtime
- Real-Time Detection

TABLE OF CONTENTS

Auto-generated.

LIST OF FIGURES

Auto-generated.

LIST OF TABLES

Auto-generated.

LIST OF ABBREVIATIONS

Include:

Abbreviation	Meaning
AI	Artificial Intelligence
CNN	Convolutional Neural Network
FER	Facial Emotion Recognition
KD	Knowledge Distillation
EMA	Exponential Moving Average
TTA	Test Time Augmentation
GUI	Graphical User Interface

ONNX	Open Neural Network Exchange
API	Application Programming Interface
CPU	Central Processing Unit
GPU	Graphics Processing Unit
UI	User Interface

CHAPTER ONE

INTRODUCTION

Target Length:

10–15 pages

1.1 Background of the Study

Writing Flow

Section 1

Human communication.

Section 2

Importance of emotions.

Section 3

Role of emotions in:

- Education
- Workplace
- Healthcare
- Human-computer interaction

Section 4

Artificial Intelligence.

Section 5

Computer Vision.

Section 6

Deep Learning.

Section 7

Facial Emotion Recognition.

Section 8

Real-time FER systems.

Section 9

Deployment challenges.

Section 10

Motivation for this study.

Figure 1.1

Evolution of FER systems.

Flow:

Traditional Methods → Machine Learning → CNN → Transfer Learning → Knowledge Distillation → Real-Time Deployment

1.2 Statement of the Problem

Writing Pattern

Current situation.

↓

Existing FER research.

↓

Existing limitations.

↓

Deployment challenges.

↓

Ethiopian context.

↓

Research need.

Mention

- High computational requirements
- Limited deployment studies
- Lack of practical desktop systems
- Lack of Ethiopian implementations
- Need for consumer hardware deployment

1.3 Research Questions

1. How can facial emotion recognition performance be improved through iterative model development?
2. What impact does transfer learning have on FER performance?
3. What impact does knowledge distillation have on deployable FER systems?
4. Which architecture provides the best balance between accuracy and deployment efficiency?

5. Can a desktop FER application operate effectively in real time on consumer hardware?

1.4 General Objective

Develop and evaluate a real-time facial emotion detection desktop application using deep learning.

1.5 Specific Objectives

Map every objective to a Chapter 4 result.

Examples:

- Aggregate and preprocess FER datasets.
- Develop baseline CNN models.
- Evaluate transfer learning architectures.
- Implement knowledge distillation.
- Develop desktop applications.
- Deploy using ONNX Runtime.
- Compare performance across model generations.

1.6 Scope of the Study

Include

Dataset scope.

Model scope.

Application scope.

Deployment scope.

Exclude

Audio emotions.

Multimodal systems.

Mobile applications.

1.7 Significance of the Study

Create four subsections.

Academic Significance

Technical Significance

Social Significance

Ethiopian Significance

1.8 Contributions of the Study

Minimum 1.5 pages.

Number them.

Contribution 1

Unified FER dataset pipeline.

Contribution 2

Five model generations.

Contribution 3

CNN architecture evaluation.

Contribution 4

Transfer learning implementation.

Contribution 5

Knowledge distillation framework.

Contribution 6

Feature distillation implementation.

Contribution 7

Three desktop application generations.

Contribution 8

YuNet integration.

Contribution 9

ONNX deployment pipeline.

Contribution 10

72.13% balanced accuracy achievement.

Contribution 11

Real-time FER deployment.

Contribution 12

Practical FER implementation in Ethiopia.

Figure 1.2

Research contribution map.

1.9 Organization of the Thesis

Briefly describe Chapters 1–5.

CHAPTER TWO

LITERATURE REVIEW

Target Length:

20–30 pages

2.1 Introduction

Explain purpose of literature review.

2.2 Emotion and Emotion Theories

Discuss:

- Ekman Theory
- Basic Emotions
- Psychological basis

Figure 2.1

Seven emotion categories.

2.3 Facial Emotion Recognition

History and evolution.

Figure 2.2

FER timeline.

2.4 Computer Vision for Face Analysis

Discuss:

- Haar Cascade
- HOG
- Dlib
- MTCNN
- RetinaFace
- YuNet

Table 2.1

Comparison of face detectors.

Figure 2.3

Face detection evolution diagram.

2.5 Deep Learning

Discuss:

- Neural networks
- CNNs
- Feature extraction

Figure 2.4

CNN architecture illustration.

2.6 Transfer Learning

Discuss:

- Concept
- Advantages
- Applications

Separate Subsections

2.6.1 MobileNetV2

Directly linked to Model V3.

2.6.2 EfficientNet Family

Directly linked to Final Model.

2.6.3 ConvNeXt

Directly linked to Teacher Network.

Figure 2.5

Transfer learning workflow.

2.7 Knowledge Distillation

One of the longest sections.

Discuss:

- Teacher model
- Student model
- Soft labels
- Temperature
- KL divergence
- Feature distillation
- Deployment benefits

Figure 2.6

Knowledge distillation architecture.

Figure 2.7

Feature distillation mechanism.

2.8 Model Optimization Techniques

Discuss:

- EMA
- TTA
- Weighted sampling
- Focal loss
- Label smoothing
- Data augmentation

Figure 2.8

Optimization pipeline.

2.9 ONNX and Model Deployment

Discuss:

- ONNX
- ONNX Runtime
- Cross-platform deployment
- Inference acceleration

Figure 2.9

ONNX deployment workflow.

2.10 Desktop FER Systems

Review practical deployment papers.

2.11 Related Works

Minimum:

15–25 papers

Table 2.2

| Author | Dataset | Method | Accuracy | Limitation |

Then analyze patterns.

Do not stop at the table.

2.12 Research Gap

Create three subsections.

Theoretical Gap

Practical Gap

Deployment Gap

Figure 2.10

Research gap diagram.

2.13 Conceptual Framework

Figure 2.11

Input Face → Detection → Preprocessing → Classification → Emotion Output

CHAPTER THREE

METHODOLOGY

Target Length:

25–35 pages

This is the most important chapter.

3.1 Introduction

Overview of methodology.

3.2 Research Design

Use:

Iterative engineering design methodology.

Figure 3.1

Research process flowchart.

Problem → Dataset → Model Development → Evaluation → Deployment

3.3 System Overview

Figure 3.2

Complete system architecture.

Webcam/Image Upload

↓

YuNet Face Detection

↓

Preprocessing

↓

ONNX Model

↓

Emotion Prediction

↓

GUI Output

This should be a professionally drawn architecture diagram.

3.4 Dataset Development

One of the longest sections.

3.4.1 Dataset Sources

Explain every dataset used in the merged dataset.

Table 3.1

| Dataset | Images | Classes |

3.4.2 Dataset Aggregation

Explain how datasets were merged.

Figure 3.3

Dataset aggregation workflow.

3.4.3 Label Standardization

Table 3.2

Emotion mapping table.

3.4.4 Dataset Cleaning

Explain:

- Duplicate removal
- Corrupt image removal
- Label verification

Figure 3.4

Data cleaning pipeline.

3.4.5 Dataset Splitting

Explain train-validation-test split.

Figure 3.5

Dataset partition chart.

3.5 Data Preprocessing

Use actual final model preprocessing.

3.5.1 Image Resizing

Teacher:

224×224

Student:

128×128

3.5.2 Normalization

ImageNet mean and standard deviation.

3.5.3 Data Augmentation

Discuss individually:

- RandomResizedCrop
- HorizontalFlip
- ColorJitter
- Rotation
- RandomErasing

Figure 3.6

Augmentation examples.

Create a montage showing one image transformed multiple ways.

3.6 Model Evolution

Unique section.

3.6.1 Model Version 1.0.0

Accuracy:

60.18%

Architecture:

Baseline CNN

Figure 3.7

V1 architecture diagram.

Include

Architecture

Training

Results

Limitations

3.6.2 Model Version 2.0.0

Accuracy:

64.10%

Improvements

Class weights

Deeper CNN

Better augmentation

Figure 3.8

V2 architecture diagram.

3.6.3 Model Version 3.0.0

Accuracy:

64.38%

Improvements

MobileNetV2

Transfer learning

Fine-tuning

Label smoothing

Figure 3.9

MobileNetV2 architecture.

3.6.4 Model Version 4.0.0

Accuracy:

60.83%

Features

Teacher-student design

Spatial attention

Multi-head architecture

Knowledge distillation

Explain why performance dropped.

Figure 3.10

V4 knowledge distillation architecture.

3.6.5 Model Version 5.0.0

Accuracy: 72.13%

Components

- ConvNeXt-Small teacher
- EfficientNetV2-S student
- WeightedRandomSampler
- Focal Loss
- Knowledge Distillation
- Feature Distillation
- EMA
- TTA
- ONNX Export

Figure 3.11 — V5 Architecture

Figure 3.12 — Teacher-Student Framework

Figure 3.13 — Feature Projector Architecture

Figure 3.14 — Training Pipeline

Discussion

- Improvements over V4
- Remaining limitations
- Motivation for developing V6

3.6.6 Model Version 6.0.0 (Final Model)

Accuracy:

[INSERT FINAL V6 ACCURACY]

This becomes the largest subsection in Chapter 3.

Components

- ConvNeXt-Small teacher
- EfficientNetV2-S student
- Knowledge Distillation
- Feature Distillation
- WeightedRandomSampler

- Focal Loss
- EMA
- TTA
- ONNX Export
- [INSERT NEW V6 IMPROVEMENTS]

Include

- Architecture
- Training strategy
- Optimization strategy
- Knowledge distillation framework
- Feature distillation framework
- Hyperparameter configuration
- Performance improvements over V5
- Deployment readiness

Figures

Figure 3.15 — V6 Final Architecture

Figure 3.16 — V6 Teacher-Student Framework

Figure 3.17 — V6 Training Pipeline

Figure 3.18 — V6 Deployment Pipeline

Discussion

- Why V6 outperformed V5
- Contribution of each new component
- Trade-off between accuracy and deployment efficiency
- Final justification for selecting V6 as the deployment model

3.7 Application Evolution

3.7.1 Desktop Application Version 1

Figure 3.15

UI Screenshot V1

Include actual application screenshot.

3.7.2 Desktop Application Version 2

Figure 3.16

UI Screenshot V2

3.7.3 Desktop Application Version 3

Figure 3.17

UI Screenshot V3

Include final deployed application UI.

3.8 Face Detection Evolution

Discuss:

- Haar Cascade
- YuNet

Table 3.3

Comparison table.

Figure 3.18

Face detector comparison.

3.9 Deployment Pipeline

Figure 3.19

Training → ONNX Export → ONNX Runtime → Desktop Application

3.10 Algorithms

Provide pseudocode.

Algorithm 1

Training process.

Algorithm 2

Knowledge distillation.

Algorithm 3

Real-time inference.

CHAPTER FOUR

RESULTS AND DISCUSSION

Target Length:

20–30 pages

4.1 Experimental Setup

Hardware

Software

Libraries

Table 4.1

System specifications.

4.2 Dataset Statistics

Figure 4.1

Class distribution bar chart.

Figure 4.2

Sample images per class.

4.3 Model Comparison

Table 4.2

Model Accuracy		Major Features
V1	60.18%	Baseline CNN
V2	64.10%	Improved CNN
V3	64.38%	MobileNetV2 + Transfer Learning
V4	60.83%	Initial Knowledge Distillation
V5	72.13%	Advanced KD Framework
V6	[FINAL ACCURACY] Final Optimized Architecture	

4.4 Training Analysis

For each model:

Figure 4.3–4.8

Training curves

Validation curves

Loss curves

for all six models.

4.5 Confusion Matrix Analysis

Show confusion matrix for final model.

Figure 4.8

Final confusion matrix.

Discuss each emotion.

4.6 Knowledge Distillation Analysis

Table 4.3

Teacher vs Student comparison.

Figure 4.9

KD framework performance gains.

4.7 Application Evaluation

Evaluate:

- Webcam performance
- Image upload performance
- Real-time inference
- Resource usage

Table 4.4

Application functionality comparison.

4.8 Deployment Analysis

Figure 4.10

ONNX deployment pipeline results.

Discuss speed improvement.

4.9 Contribution Validation

Create one subsection per contribution.

For each:

Problem

Solution

Result

Impact

4.10 Discussion

Explain:

- ❓ Why V1 achieved 60.18%
- ❓ Why V2 improved to 64.10%
- ❓ Why V3 improved to 64.38%
- ❓ Why V4 decreased to 60.83%

- ❓ Why V5 improved to 72.13%
- ❓ Why V6 achieved the highest performance
- ❓ Comparative analysis of all six generations
- ❓ Final architecture selection rationale

This section demonstrates research maturity.

CHAPTER FIVE

CONCLUSION AND FUTURE WORK

Target Length:

6–10 pages

5.1 Conclusion

Structure:

Problem

Methodology

Results

Contributions

Deployment

5.2 Limitations

Include:

- Occlusions
- Lighting
- Accessories
- Dataset imbalance
- Single-modality FER
- Domain shift

5.3 Recommendations

Academic recommendations.

Technical recommendations.

Deployment recommendations.

5.4 Future Work

Include:

- Vision Transformers
- Multimodal FER
- Audio-visual FER

- Mobile deployment
- Cloud deployment
- Edge AI
- Ethiopian FER dataset development
- Multilingual emotion analytics

REFERENCES

Target

70–100 references

Distribution

- 60% from 2020–2026
- IEEE papers
- Springer papers
- Elsevier papers
- ACM papers

Must Cover

- FER
- CNN
- MobileNetV2
- EfficientNetV2
- ConvNeXt
- Knowledge Distillation
- Feature Distillation
- ONNX Runtime
- YuNet
- Face Detection
- Human Emotion Analysis

APPENDICES

APPENDIX A — Dataset Documentation

Include:

- Dataset sources
- Dataset statistics
- Class distributions

Figures

Additional dataset charts.

APPENDIX B — Model Architectures

Include full architecture summaries for:

- V1
- V2
- V3
- V4
- V5
- V6

Use exported model summaries.

APPENDIX C — Application Interfaces

Place all GUI screenshots here **and also include the final application UI once in Chapter 3 (Application Evolution) and once in Appendix C for full reference.**

Include:

- Version 1 UI
 - Version 2 UI
 - Version 3 UI
 - Webcam mode
 - Image upload mode
 - Detection results screen
 - Logging screen (if available)
-

APPENDIX D — Experimental Results

Include:

- Additional confusion matrices
 - Additional performance tables
 - Training logs
 - Hyperparameter tables
-

APPENDIX E — Algorithms and Pseudocode

Include full versions of:

- Training algorithm
- Knowledge distillation algorithm
- Real-time inference algorithm

APPENDIX F — Source Code Excerpts

Do **not** paste entire codebases.

Only include important snippets:

- ConvNeXt teacher initialization
- EfficientNetV2 student initialization
- Knowledge distillation loss
- ONNX export
- ONNX Runtime inference
- YuNet integration

APPENDIX G — User Manual

Step-by-step:

1. Install application.
2. Launch application.
3. Load image.
4. Start webcam.
5. View predictions.
6. Read logs.
7. Exit application.

Include screenshots.

APPENDIX H — Project Publications and Outputs (Optional)

Include:

- Conference submissions
- Posters
- Presentations
- Demonstration materials

WRITING RULES FOR EVERY CHAPTER

Paragraph Structure

Use:

Claim → Evidence → Explanation → Link

Never write one-sentence paragraphs.

Figure Rule

Every figure must:

1. Be introduced before it appears.
2. Be explained after it appears.
3. Be referenced in text.

Example:

"Figure 3.11 illustrates the final teacher-student architecture used in the proposed system. The architecture combines ConvNeXt-Small as the teacher network and EfficientNetV2-S as the student network. This design enables effective knowledge transfer while maintaining deployment efficiency."

Table Rule

Never insert a table without analysis.

Always follow:

Table → Interpretation → Discussion.

Result Discussion Rule

Never write:

"The accuracy was 72.13%."

Write:

"The final model achieved a balanced accuracy of 72.13%, representing a substantial improvement over the baseline CNN architecture. This improvement can be attributed to the integration of transfer learning, feature-level knowledge distillation, weighted sampling, and exponential moving average optimization."

Thesis Positioning Rule

Throughout Chapters 1, 3, and 4, consistently present the work as:

"An iterative design, optimization, deployment, and evaluation framework for real-time facial emotion recognition"

rather than simply:

"A facial emotion detection application."

You're welcome.

One final piece of advice for maximizing your thesis grade:

Focus your effort in this order

1. **Chapter 3 (Methodology)** — 30%
2. **Chapter 4 (Results & Discussion)** — 25%
3. **Chapter 2 (Literature Review)** — 20%
4. **Chapter 1 (Introduction)** — 15%
5. **Chapter 5 (Conclusion)** — 10%

For your specific project, Chapters 3 and 4 are where you can separate yourselves from most BSc projects because you have:

- 5 model generations (V1–V5)
- 3 desktop application generations
- Dataset aggregation
- CNN experiments
- Transfer learning
- Knowledge distillation
- Feature distillation
- ONNX deployment
- YuNet integration
- Real-time desktop deployment
- Final 72.13% model

Many students only have one model and one result table. You have an entire development lifecycle to document.

What will impress examiners most

- Clear justification for each model evolution
- Explanation of why V4 decreased before V5 improved
- Strong research gap section
- Strong contribution section
- Detailed methodology diagrams
- Thorough comparison table of all versions
- Professional architecture figures
- Good discussion of deployment and practical usability
- Recent references (2020–2026)

Biggest mistakes to avoid

- ✗ Writing Chapter 2 as paper summaries one after another
- ✗ Describing code line-by-line
- ✗ Putting screenshots without explanation
- ✗ Reporting accuracy without analysis
- ✗ Claiming novelty without evidence
- ✗ Mixing implementation details into the literature review

✗ Using fewer than ~50 references

Strong contribution wording

Instead of:

"We built an emotion detection system."

Use:

"This study presents the iterative design, optimization, deployment, and evaluation of a real-time facial emotion recognition system, progressing from baseline CNN architectures to a knowledge-distilled ONNX-deployed solution suitable for consumer hardware environments."

That wording reflects the actual scope of your project and sounds much more like research.

Based on everything you've shared in this chat (project title, objectives, model versions V1–V5, application versions V1–V3, deployment pipeline, and thesis expectations), the blueprint above is comprehensive enough to serve as the master structure for writing the full BSc thesis from the cover page through the appendices.